

ABSTRACT

The conventional paddy farming system in Tamil Nadu is largely based on visual inspection and uniform irrigation that cause poor utilization of water resources and lack of information about the root zone. It involves high labour intensity and ignores field heterogeneity. As water scarcity becomes increasingly pressing, efficient utilization of available resources is becoming critical. Thus, a necessity emerges to design intelligent and automated irrigation systems capable of ensuring sustainable agricultural production. The design of the Agrimind, a Mobile Iot-Based Robotic System is intended to monitor soil conditions and irrigate fields with respect to them. The core of the system is a rover equipped with various sensors measuring soil moisture, pH, temperature, humidity, and pressure at specific points within the field. In contrast to static systems, the proposed design allows for mobile measurement of soil properties. The novelty of the developed solution lies in the bio-impedance sensing method implemented in the rover. Using bio-impedance, the system analyses the state of roots by examining the impedance of root-soil interfaces. The rover measures impedance parameters

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ABBREVIATIONS

1. CCU – Central Control Unit
2. IoT – Internet of Things
3. GPS – Global Positioning System
4. UART – Universal Asynchronous Receiver/Transmitter
5. I2C – Inter-Integrated Circuit
6. SPI – Serial Peripheral Interface
7. ADC – Analog to Digital Converter
8. ESP – Espressif Systems Microcontroller
9. BMP – Barometric Pressure Sensor
10. PWM – Pulse Width Modulation
11. HTTP – Hyper Text Transfer Protocol
12. MQTT – Message Queuing Telemetry Transport

CHAPTER I

INTRODUCTION

1.1 Objectives

The main aim of this AGRIMIND project is to design a smart and mobile IoT-based system for soil sensing that can be effectively used for the purpose of monitoring the soil condition in paddy fields. With the help of this system, farmers will be able to take appropriate actions with regard to the irrigation of their crops based on the soil and environmental conditions. A key objective of the system is to design and implement a mobile robotic rover capable of operating effectively in challenging agricultural conditions such as wet, muddy, and uneven paddy fields. The rover is intended to move across different regions of the field and collect localized data, thereby overcoming the limitations of stationary monitoring systems. This mobility ensures better coverage and more accurate representation of field conditions.

Another vital aim is the combination of various sensors for the determination of several crucial values like soil moisture content, pH value, temperature, and air pressure. These values are crucial for assessing the condition of soil and its productivity. The proposed system will be capable of collecting and analysing the data continuously with the help of the ESP32 microcontroller that will serve as the main controller of the system. The project also focuses on enabling wireless communication using Wi-Fi, allowing the collected data to be transmitted to external devices or cloud platforms. This feature supports remote monitoring, enabling farmers to access field data without being physically present. In addition, the integration of a GPS module allows location-based data tagging, which helps in mapping different zones of the field and analyzing spatial variations in soil conditions.

Additionally, the system intends to facilitate proper irrigation management by analyzing soil moisture content and providing information regarding the need for water. This will play an important role in conserving water resources through improved water management. Moreover, the purpose of this project is to cut down on labor efforts as much as possible by incorporating automation features into data gathering and monitoring procedures.

Finally, the AGRIMIND system intends to be affordable, flexible, and adaptable to various types of agricultural settings. The main goal is to build a sustainable and intelligent system that boosts efficiency and facilitates precision agriculture.

2. Brief Description of project

AGRIMIND is an Internet of Things based mobile system that is used to monitor soil health during rice cultivation process. The system utilizes embedded systems, sensors, and wireless communication technology to facilitate monitoring of soil and environment properties. This innovation addresses inefficiencies associated with traditional farming where farmers use manual monitoring and irrigation processes that do not yield high levels of efficiency.

The main component of this innovative solution is an ESP32-based mobile robot that can move around the farm and monitor different areas of the farm using sensors. The robot has sensors such as capacitive soil moisture sensor, pH sensor, and BMP280 environmental sensor, among others. Sensors enable measurement of important properties such as soil moisture, pH level, soil temperature, and atmospheric pressure, among others. A unique feature of the system is its ability to perform localized sensing at different zones of the field, which helps in identifying variations in soil conditions. The integration of a GPS module enables location-based data collection, allowing the system to map soil properties across the field. This supports zone-wise analysis and helps in making precise agricultural decisions. The data collected from sensors is processed by the ESP32 microcontroller and transmitted wirelessly using Wi-Fi for remote monitoring.

It is also equipped with motor controlling mechanisms with the help of L298N drivers so that the robot can move around the field. Ultrasonic sensors have been provided for detecting obstacles to avoid any accident while traversing.

The recorded data can also be monitored on any external device or even from cloud platforms.

3. Problem Statement

The traditional techniques used for paddy farming involve the process of visual inspection of the fields and even uniform irrigation of the crop, which ultimately leads to poor water utilization and inadequate knowledge about soil characteristics. Farmers usually base their knowledge about the necessity of irrigation on experience and visual assessment of the farm; however, this may not always be an accurate depiction of the true state of the soil in terms of its water requirement.

One of the significant issues that the traditional technique does not solve is providing real-time knowledge about the soil parameters, which include moisture content and pH levels. It should be noted that different parts of a land have different soil qualities. Unfortunately, traditional techniques

are unable to provide information about individual zones, making the process of irrigation and soil treatment highly ineffective.

Moreover, manual monitoring of vast agricultural areas requires effort and time. The farmers have to manually check different sections of the field, which makes the process even more difficult and might not help in identifying any stress or imbalance in plants or soil. Environmental parameters like temperature and humidity also affect the development of the crops; however, these factors are not continuously being monitored using conventional systems.

The current methods of agriculture do not include the process of collecting data, analysing it, and taking any decisions accordingly, as no automation tool has been used to do so. Without having any automatic tool, farmers cannot make proper decisions using the collected data since it has to be analysed first.

Consequently, there is a need for developing a smart and automated system that will continuously monitor different factors affecting the field soil and environment and also provide real-time data to make decisions regarding water irrigation. Such a tool must be able to work effectively under tough paddy field conditions and be useful in increasing the yield.

1.4 Existing Systems

For traditional paddy farming, farmers depend largely on visual examination and conventional irrigation methods in determining and managing soil conditions. These include physically assessing soil moisture, plant development, and water requirement in the field. Conventional irrigation techniques also apply the same method of uniform irrigation all throughout the field without factoring in the variation of soil conditions. The consequence is water wastage, poor crop development, and lowered productivity. Manual monitoring methods are also costly, tedious, and laborious to undertake consistently within vast agricultural land areas.

Technology has provided ways through which agriculture can be efficiently undertaken using several innovations. One notable innovation is the FarmBot Genesis, an open-source precision agriculture tool that utilizes the CNC technique for automatic planting, watering, and monitoring. It works in an enclosed area wherein sensors determine soil conditions to implement programmed actions. Despite providing efficient control and automation services, the FarmBot Genesis operates in a confined,

defined, and fixed area such as garden farming. This makes it suitable for limited areas but impractical when applied to extensive and open agricultural land such as paddy farming.

In other agriculture systems using IoT, stationary sensors are used to measure soil moisture levels, temperature, and humidity. This type of agriculture system transmits its data into cloud computing platforms. Although it can offer real-time data for analysis, it does not offer mobility and thus fails to capture any spatial changes occurring on the field. It does not fully address the issue of zone-wise monitoring and control of irrigation processes.

Moreover, most systems do not emphasize root-zone analysis for the health of plants. Most systems also lack a proper relationship between sensing, mobility, and decision-making process which limits their performance in real-life situations. There is also no location-based monitoring and automatic field coverage for proper analysis.

In conclusion, although various systems can be used for agricultural automation including the conventional method, stationary IoT sensors, and even the FarmBot Genesis platform, there exist some limitations with these systems including scalability, mobility, and applicability in paddy fields.

1.5 Proposed System

The idea behind designing the new system to counteract the shortcomings of previous technologies is to provide a mobile system using Internet of Things (IoT) technology for paddy fields. The new system combines several functionalities such as sensing, processing, mobility, and communication within itself. Unlike static systems, where data about a specific area only is available, the proposed design uses a mobile robot to traverse the entire field and gather data from different regions.

The main component of this system is a central control unit based on ESP32 microcontrollers, which controls all activities, from sensor connectivity to motor control and even wireless communication. The robot has been designed with several sensors, including the capacitive soil moisture sensor, pH sensor, and BMP280 environment sensor, which can detect the parameters of soil moisture, soil acidity, temperature, and air pressure. All these factors are very important in determining the health status of the soil.

Another important aspect of the suggested system is that it enables localized sensing in the root zone area in various parts of the field. With the help of a GPS unit, the system will be able to attach the

geographic location information to the data collected by the sensors. Thus, the system will facilitate the mapping of the soil in various zones and help farmers plan the required irrigation and soil management strategy for each individual zone, which will lead to more efficient use of resources and higher crop yield.

Controlling the movements of the roving robot is done via the use of a motor driver module, which will facilitate the navigation of the field to collect data continuously. The data collected can be sent to other devices wirelessly using Wi-Fi technology. This minimizes the need for physical inspection by the farmer who can make sound decisions using the data.

In summary, the proposed AGRIMIND system presents itself as an efficient, scalable, and intelligent means of achieving precision agriculture. The integration of IoT and Robotics technology, together with GPS tracking in the system, leads to improved water usage, reduced need for labour, and increased crop yield. It is able to overcome the shortcomings of current agricultural systems.

6. Technologies used in Proposed Systems

1. Hardware Platforms

- 1. Central Control Unit (CCU):** It is the core processor of the AGRIMIND system, which is based on the ESP32 microcontroller. It includes the necessary communication modules and sensors to gather and process real-time data related to the soil and environment. All the activities of the rover, such as sensing, data processing, and communication, are managed by the CCU.
- 2. Mobile Robotic Rover Platform:** It is the physical entity that travels through the paddy field and gathers information. This rover has motors, wheels, and a motor driver (L298N). The purpose behind including them is to move on muddy or uneven terrain. This rover enables us to monitor in zones.
- 3. Sensor Modules:** The system employs several sensors for measuring various parameters, such as the capacitive soil moisture sensor, used to measure moisture levels; the pH sensor, used to measure the level of soil acidity; and the BMP280 sensor, which measures both temperature and atmospheric pressure.

4. **GPS Module:** The GPS module is used for positioning and mapping purposes. It provides the ability to position the sensors' data according to geographical positions, which can be used for zone-wise analysis of the soil.
5. **Wireless Communication (Wi-Fi):** The data gathered by the sensors on the rover can be transmitted using Wi-Fi communication. This is made possible by the onboard Wi-Fi module in the ESP32.
6. **Power Supply:** This is done through the use of rechargeable batteries for powering the system. Voltage regulators are implemented to ensure constant supply of electricity to the ESP32, sensors, and motor controller.

1.6.2 Software Frameworks & Libraries

1. **Embedded C/C++ Programming:** The embedded C/C++ language is utilized to program the ESP32 microcontroller. It offers precise hardware component management, including sensor management, communication devices, and motors, among others, enabling effective and efficient performance of operations.
2. **Arduino / ESP-IDF Frameworks:** Such development platforms include Arduino IDE and ESP-IDF, which help program the ESP32 microcontroller. Development platforms help in developing systems faster since they make programming easier by reducing complications when connecting to the hardware components.
3. **Sensor Libraries:** The Adafruit BMP280 library is an example of a specialized library that is employed to communicate with environmental sensors. Such libraries enable communication with the microcontroller through dedicated functions for data acquisition.
4. **Communication Libraries:** WiFi-related libraries like WiFi.h are useful in creating a wireless connection and handling communication between devices. They help in connecting to the internet and communication protocol implementation for monitoring purposes.

5. **GPS Parsing Libraries:** Libraries like TinyGPS++ are helpful in processing the GPS data to find locations such as latitude and longitude. With these data, we can map the location of soils accurately.
6. **Cloud Communication Protocols:** Lightweight communication protocols such as HTTP and MQTT can be employed for data transmission to cloud services. The protocols guarantee the transfer of data in an effective and consistent manner.
7. **Artificial Intelligence (Future Enhancement):** Machine learning frameworks such as TensorFlow Lite and Edge Impulse can be employed to incorporate features such as crop monitoring and object recognition using AI.

1.6.3 Programming Languages

C/C++ programming language is mainly intended for coding the Central Control Unit (CCU) that will be designed based on the ESP32 microcontroller to be used in the AGRIMIND framework. This programming language allows managing the hardware directly and provides an opportunity to handle data acquired from different sensors, transmit the data via WiFi networks, and interact with other elements like soil moisture sensors, pH sensors, BMP280 board, GPS board, and motor driver L298N.

1.6.4 Development Tools

1. **Arduino IDE:** Platform that is convenient to use for programming ESP32 processors. It can be programmed using C/C++. The libraries for sensors, Wi-Fi, GPS, and motors are included in it.
2. **ESP-IDF (Espressif IoT Development Framework):** ESP-IDF can be utilized for sophisticated programming purposes on ESP32 boards. It offers enhanced capabilities for controlling hardware resources, wireless communication, and peripherals. It is well-suited for creating highly efficient embedded software systems.
3. **Visual Studio Code (VS Code):** Code editor which can be used to write and manage embedded code written using C/C++. It can be enhanced with plugins such as PlatformIO and makes it easy to develop, debug, and maintain libraries for an ESP32 project.

4. **Proteus / Multisim:** Simulation software which is used during circuit design and verification prior to hardware development. They allow visualization of connections and behavior of circuits to diagnose problems related to hardware development.
5. **Git and GitHub:** Version control software used to manage and document various aspects of code development within the project. They allow maintenance of different versions of embedded code.
6. **Cloud Platforms (Firebase / AWS / Google Cloud):** The cloud platforms can be used to store, manage, and visualize the collected information from the sensors. The cloud platforms allow the real-time monitoring and analysing the data, as well as remote access to it via a webbased or mobile interface.
7. **Serial Monitor Tools:** Debugging tools like the Arduino Serial Monitor can be employed during testing and debugging phases of the project development.

1.6.5 Communication Protocols

1. Wi-Fi Communication (IEEE 802.11)
2. HTTP Protocol
3. MQTT Protocol (Optional)
4. I2C Communication Protocol
5. Analog-to-Digital Conversion (ADC)
6. UART Communication (GPS Module)

CHAPTER II

LITERATURE SURVEY

The recent innovations in smart agriculture are centred on the application of IoT technologies to monitor the situation on the farm in real-time and manage the available resources efficiently. Research, including [1] and [6] points to the fact that IoT allows gathering consistent data and making more informed decisions in agriculture. The merger of big data and cloud computing is another factor helping to increase the productivity and sustainability of agriculture [2], [10].

A major contribution is made in the article of Smart Saffron Cultivation: An IoT-Based Environmental Monitoring and Control System with GSM Alert Integration [4]. This system measures environmental conditions, including temperature, humidity and soil moisture using sensors and sends the data through GSM. It gives notifications to the farmers, allowing them to take actions in time and manage their crops better. Other systems based on GSM/GPRS communication have been discussed in [3], where energy-efficient monitoring is considered.

But the majority of systems present are fixed and are unable to record the changes in the various regions of the field. They are also dependent on GSM communication which can create delays and it does not offer high speed continuous transmission of data. Also, other systems are confined to structured settings such as FarmBot Genesis [12] and agricultural robotics platforms [11], which do not work on large, unstructured fields such as paddy cultivation.

To overcome these shortcomings, the proposed AGRIMIND system proposes a mobile IoT-based platform that can be used to collect data zone-wise. It combines GPS to track their location, Wi-Fi to communicate in real-time, and multi-parameter sensing to be more accurate. Therefore, it complements current methods by offering superior coverage, scalability and accuracy in monitoring agriculture.

CHAPTER III

METHODOLOGY

1. Hardware design

1. Central Control Unit (CCU)

1. Microcontroller (ESP32)

ESP32 microcontroller has been chosen because of its high speed, low power usage, and builtin wireless connectivity. This device uses dual cores, allowing it to simultaneously perform several functions such as sensing data collection and communication tasks. The ESP32 collects the input from the sensors, analyses the information, and produces corresponding control commands for the motor control and alert systems.

The Wi-Fi feature built into the ESP32 will help send collected data to the Internet of Things platform or an app. Moreover, the availability of many GPIO pins makes it easy to connect several modules and sensors.

2. Soil Moisture Sensor

The function of the soil moisture sensor is to detect the amount of water present within the soil. This works through the measurement of the resistance or electrical conductivity that changes with the change in moisture. The data output is then converted from analog to digital format by the ESP32.

This data will be used to identify whether or not there is a need for irrigation of the soil. This data will aid in avoiding either under- or over-watering the soil, which will help in saving water and improve the crops' output. The sensor should work well outdoors, but calibration will be needed.

3. pH Sensor

The pH sensor measures the acid or base nature of the soil, an important parameter affecting the availability of nutrients for crops. The pH sensor sends an analog signal that indicates the pH level ranging from 0 to 14. The ESP32 analyzes the data and identifies the soil as being acidic, neutral, or alkaline.

Proper measurement of pH will allow farmers to determine which fertilizer or soil treatments should be done. pH sensors being used are very sensitive hence need for frequent calibration with standard solutions.

4. Environmental Sensor (BMP280)

BMP280 is the name of the sensor that will be used to measure temperature and pressure in the atmosphere. It is important to note that these environmental factors play a critical role in the rate at which moisture evaporates from the soil and the health of plants.

Due to its accuracy and minimal power consumption, the BMP280 is well suited for monitoring the environment continuously. This data can also be applied for predictive analyses, including forecasting trends in the weather on agricultural land.

5. Motor Driver (L298N)

L298N motor driver is employed to control the motion of the rover. This motor driver facilitates the two-directional control of the DC motor by ensuring that the robot is able to move forwards, backwards, and can turn.

The L298N motor driver is commonly used because it is easy to use and readily available. However, the L298N is less efficient since there is more heat dissipation. However, it is adequate for use in testing.

6. Power Supply Circuit

The power supply unit supplies stable voltage to all the modules. This module regulates the voltage from the battery for various modules, like 3.3V for ESP32 and 5V for the sensor and motor driver.

Safety features like voltage regulation, filtration, and short-circuit protection are provided in order to provide safe operations. The system is constructed in such a way that it functions effectively under varying levels of electrical supply present in practical environments.

2. Mobility Unit

The mobility unit enables the rover to traverse agricultural fields efficiently. It is designed considering the challenges posed by muddy and uneven terrain.

1. BO Motors

BO motors are used for driving the wheels of the rover. These motors are lightweight, cost-effective, and provide sufficient torque for movement across soft soil surfaces.

2. Wheels (Lugged Agricultural Type)

The wheels are specially designed with deep treads to provide better grip in wet and muddy conditions. This prevents slipping and ensures smooth movement of the rover in paddy fields.

3. Communication Interface

1. Wi-Fi Communication

Communication through Wi-Fi is employed for sending the sensor data to other devices, including smartphones and cloud servers. It allows real-time monitoring and remote access to the field data. All communication protocols are managed efficiently by the ESP32.

3.2 Software Design

The software is designed in a modular way to ensure smooth operation and easy maintenance. Each module handles a specific function such as sensing, processing, or communication.

1. Central Control Software

1. Sensor Data Acquisition Module

This module reads data from all sensors continuously and updates the system.

2. Data Processing Module

Processes raw data and converts it into meaningful values.

3. Motor Control Module

Controls rover movement based on system logic.

4. Communication Module

Handles Wi-Fi communication and data transmission.

5. Alert System

Generates alerts when sensor values exceed limits.

CHAPTER IV

IMPLEMENTATION

1. Hardware Module 1.

Microcontroller (ESP32)

1. Working:

- Acts as the central control unit of the system
- Initializes and interfaces with all sensors
- Reads analog and digital data from sensors
- Processes sensor values into meaningful information
- Sends control signals to motor driver
- Handles Wi-Fi communication for data transmission

2. Specifications:

- Processor: Dual-core Tensilica Xtensa LX6 microprocessor (for ESP32).
- Operating Voltage: 3.3V.
- Flash Memory: 4 MB (expandable).
- RAM: 520 KB SRAM.
- Communication Interfaces: UART, SPI, I2C, CAN (for vehicle ECU communication).
- Connectivity: Wi-Fi and Bluetooth (optional for future expansion).
- Power Consumption: Ultra-low power mode supported (~10 μ A deep sleep).



Fig 4.1 Microcontroller (ESP32)

4.1.2 Soil Moisture Sensor

4.1.2.1 Working

- Detects moisture content in soil
- Outputs analog signal based on water level
- ESP32 reads and converts it into percentage value
- Helps determine irrigation requirement

4.1.2.2 Specifications

- Operating Voltage: 3.3V–5V
- Output Type: Analog
- Measurement Range: 0% to 100% moisture
- Response Time: Fast (within milliseconds)



Fig 4.2 Capacitive Soil Moisture Sensor

4.1.3 pH Sensor

4.1.3.1 Working

- Measures acidity or alkalinity of soil
- Produces analog voltage corresponding to pH value
- ESP32 converts voltage into pH reading
- Used for soil quality analysis.

4.1.3.2 Specifications

- Measurement Range: pH 0 to 14
- Accuracy: ± 0.1 pH
- Output: Analog voltage
- Calibration: Required using buffer solutions



Fig 4.3 Soil pH Sensor Probe with Signal Conditioning Module

4. BMP280 Digital Temperature and Pressure Sensor Module

1. Working

- Measures temperature and atmospheric pressure
- Sends digital data to ESP32
- Uses I2C/SPI communication

- Helps monitor environmental conditions

4.1.4.2 Specifications

- Voltage: 1.8V–3.6V
- Interface: I2C / SPI
- Temperature Range: -40°C to +85°C
- Pressure Range: 300–1100 hPa
- High accuracy



Fig 4.4 BMP280 Digital Temperature and Pressure Sensor Module

4.1.5 GPS Module (Neo-6M GPS Receiver Module)

4.1.5.1 Working

- Receives signals from multiple GPS satellites
- Determines geographical location using triangulation
- Provides latitude and longitude coordinates
- Sends data to ESP32 via UART communication
- Enables location tracking of the rover
- Can be used for future autonomous navigation and mapping

4.1.5.2 Specifications

- Module: Neo-6M GPS
- Voltage: 3V–5V

- Communication: UART
- Position Accuracy: ~2.5 meters
- Update Rate: 1 Hz (adjustable)
- Channels: 50-channel receiver
- Time to First Fix:
- Cold Start: ~27 sec
- Antenna: External ceramic antenna
- Backup Battery: Supports faster satellite acquisition



Fig 4.5 Neo-6M GPS Receiver Module

6. L298N Dual H-Bridge Motor Driver Module 1. Working

- Receives control signals from ESP32
- Drives motors using H-bridge configuration
- Enables forward and backward motion
- Controls speed using PWM signals
- Allows turning and navigation

2. Specifications

- Voltage: 5V–35V
- Current: up to 2A per channel
- Control: PWM
- Dual motor control
- Heat sink required



Fig 4.6 L298N Dual H-Bridge Motor Driver Module

7. DC Power Supply Module 1. Working

- Supplies power to all components
- Converts battery voltage to required levels
- Maintains stable voltage output
- Protects system from fluctuations

2. Specifications

- Input: 7.4V–12V battery
- Output: 5V, 3.3V
- Protection: Overvoltage, short circuit
- Efficient regulation



Fig 4.7 Power Supply Unit

4.1.8 Connectors and Wiring Harness

4.1.8.1 Working

- Connects all modules together
- Ensures proper signal transmission
- Maintains system reliability

4.1.8.2 Specifications

- Connector Type: JST, Molex, or automotive-grade clips.
- Wire Gauge: 22 AWG (Signal), 16–18 AWG (Power).
- Features: Heat resistant, abrasion proof, flexible.



Fig 4.8 Connectors and Wiring Harness

2. Software Model

1. Software Architecture Overview

The software architecture is divided into multiple functional layers to improve efficiency and maintainability.

The main layers include:

- Sensor Data Handling Layer
- Control Logic Layer
- Communication Layer
- User Interface Layer

1. Sensor Data Handling Layer

This layer is responsible for collecting data from all sensors connected to the system. It ensures that the sensor values are updated continuously and accurately.

- Reading data from soil moisture sensor, pH sensor, and BMP280
- Converting analog signals into digital values

- Filtering noise from sensor readings
- Updating real-time data continuously

2. Control Logic Layer

The control logic layer processes the sensor data and makes decisions based on predefined conditions. It plays a key role in determining system behaviour.

- Analyses soil moisture and pH values
- Compares values with pre-defined thresholds
- Determines soil condition (dry, optimal, or wet)
- Controls rover movement based on logic
- Generates alerts if abnormal conditions are detected.

3. Communication Layer

The communication layer handles the transmission of processed data from the ESP32 to external devices. It ensures reliable and real-time data transfer.

- Establishing Wi-Fi connection using ESP32
- Formatting sensor data for transmission
- Sending data to mobile devices or cloud platforms
- Maintaining continuous communication

4. User Interface Layer

This layer provides interaction between the system and the user. It allows the user to monitor realtime data remotely.

- Displaying sensor values on mobile or cloud dashboard
- Providing real-time updates
- Enabling remote monitoring of field conditions

2. Software Flow

The software flow within the system is divided into key stages:

1. Initialization

Initialization is the first stage of the software execution where all system components are prepared for operation. When the power supply is turned ON, the ESP32 begins its setup process and configures all connected peripherals.

During this stage, the following operations are performed:

- ESP32 microcontroller is powered and reset
- GPIO pins are configured for sensors and motor driver
- Communication protocols such as I2C (for BMP280) are initialized
- ADC channels are configured for analog sensors (moisture and pH)
- Serial communication is started for debugging
- Wi-Fi module is initialized and attempts to connect to network

Once all modules are successfully initialized, the system moves to the next stage.

2. Data Collection

After initialization, the system enters the data collection stage where all sensors begin to capture realtime data from the environment and soil. This stage ensures continuous monitoring of field conditions.

- Soil moisture sensor measures water content in soil
- pH sensor detects acidity or alkalinity of soil
- BMP280 sensor measures temperature and pressure
- Ultrasonic sensor checks for nearby obstacles
- Sensor values are read periodically by ESP32

The collected data is stored temporarily in variables for further processing.

3. Data Processing

In this stage, the raw sensor data is processed and converted into meaningful information. Since sensor outputs may contain noise or variations, proper processing ensures accuracy.

- Converting analog signals into digital values using ADC
- Applying calibration formulas for pH and moisture
- Filtering noise and stabilizing sensor readings
- Converting values into standard units (percentage, pH value, °C)
- Storing processed data for decision-making

This step ensures that the system works with accurate and usable data.

4. Decision Making

The decision-making stage is where the system analyses the processed data and determines appropriate actions. It is based on predefined threshold values and logic conditions.

- Compares soil moisture with threshold levels (dry / optimal / wet)
- Evaluates pH value for soil condition
- Checks environmental conditions using BMP280 data
- Determines whether rover should move or stop
- Decides if alerts need to be generated

5. Control and Execution

Based on the decisions made, the system executes control actions. This mainly involves controlling the rover movement and system outputs.

- Sending signals to L298N motor driver
- Controlling motor direction (forward, backward, turn)
- Adjusting speed using PWM signals
- Activating indicators (LEDs/buzzer if used)

This stage ensures that the system responds dynamically to environmental conditions.

6. Data Transmission

Once the system processes the data and performs actions, the information is transmitted to external devices. This enables remote monitoring of field conditions.

- Formatting sensor data into readable format
- Sending data via Wi-Fi using ESP32
- Updating mobile app or cloud dashboard
- Ensuring continuous communication

This allows the user to access real-time data from anywhere.

7. Continuous Loop Operation

The system operates continuously in a loop to provide real-time monitoring and control. After completing one cycle, the system repeats all steps again.

- Re-reading sensor data
- Re-processing values

- Re-evaluating conditions
- Updating outputs and communication

3. Software Modules and Libraries

1. Libraries

The following libraries are used in the system for efficient implementation

- **WiFi.h:** This library is used to enable Wi-Fi communication in the ESP32. It helps in connecting the device to a wireless network and transmitting sensor data to external devices or cloud platforms.
- **Wire.h:** The Wire library is used for I2C communication. It allows the ESP32 to communicate with sensors like BMP280 using two-wire communication (SDA and SCL).
- **Adafruit BMP280 Library:** This library provides functions to initialize and read data from the BMP280 sensor. It simplifies temperature and pressure data acquisition.
- **Arduino Core Library:** This includes built-in functions provided by the Arduino framework for basic operations such as GPIO control, ADC reading, and timing.

2. Functions and Procedures

- **Initialization Function:** This function is executed at the start of the program. It initializes all hardware components such as sensors, motor driver, and communication modules. It also sets pin modes and establishes Wi-Fi connection.
- **Sensor Reading Function:** This function reads data from soil moisture sensor, pH sensor, and BMP280. It collects raw data and updates the system continuously.
- **Data Processing Function:** This function processes raw sensor values, applies calibration, and converts them into meaningful units such as percentage, pH value, and temperature.
- **Control Function:** This function analyzes processed data and makes decisions based on predefined thresholds. It controls rover movement and system behaviour accordingly.
- **Motor Control Procedure:** This procedure sends signals to the L298N motor driver to control motor direction and speed. It ensures smooth navigation of the rover.
- **Communication Function:** This function handles Wi-Fi data transmission. It sends processed sensor data to external devices or cloud platforms for monitoring.
- **Loop Procedure:** This is the main execution loop of the program. It continuously performs sensing, processing, decision-making, and communication operations in a cycle.

3. Arduino (ESP32) Code

1. Central Control Unit (Rover) Code:

```
#include <WiFi.h>
#include <Wire.h>
#include <Adafruit_BMP280.h>

// Sensor Pins const int moisturePin = 34;
// Analog pin const int phPin = 35;
// Analog pin

// Motor Driver Pins
(L298N) const int ENA = 12;
const int IN1 = 13; const int
IN2 = 15;

// Wi-Fi Credentials const char* ssid =
"AgriMind_Network"; const char*
password = "12345678";

// BMP280 Sensor
Adafruit_BMP280 bmp;

// Variables int
moistureValue = 0; int
phValue = 0; float
temperature = 0; float
pressure = 0;
WiFiServer
server(80);

void setup() {
  Serial.begin(115200);
```

```

// Initialize Motor Pins
pinMode(ENA,
OUTPUT); pinMode(IN1,
OUTPUT); pinMode(IN2,
OUTPUT);

// Stop motor initially
analogWrite(ENA, 0);
digitalWrite(IN1,
LOW);
digitalWrite(IN2,
LOW);

// Initialize Sensors
Wire.begin();
bmp.begin(0x76);

// Connect to Wi-Fi
WiFi.begin(ssid, password);
Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nConnected!");
Serial.println(WiFi.localIP());

server.begin();
}

void loop() {

```

```

// Read Sensors moistureValue =
analogRead(moisturePin); phValue =
analogRead(phPin); temperature =
bmp.readTemperature(); pressure =
bmp.readPressure() / 100.0;

// Simple Decision Logic if
(moistureValue < 2000) { //
Move forward (dry soil)
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
analogWrite(ENA, 150);
} else {
// Stop (sufficient moisture) analogWrite(ENA,
0);
}

// Print Data
Serial.print("Moisture: ");
Serial.print(moistureValue);
Serial.print(" | pH: ");
Serial.print(phValue);
Serial.print(" | Temp: ");
Serial.print(temperature);
Serial.print(" °C | Pressure: ");
Serial.print(pressure);
Serial.println(" hPa");

// Send Data via Wi-Fi WiFiClient
client = server.available(); if (client) {
client.println("HTTP/1.1 200 OK");
client.println("Content-
type:text/html"); client.println();

```



```

    client.print("Moisture: ");
    client.print(moistureValue);
    client.print("<br>pH: ");
    client.print(phValue);
    client.print("<br>Temperature: ");
    client.print(temperature);
    client.print("<br>Pressure: ");
    client.print(pressure);

    client.stop();
}

delay(2000);
}

```

4.3.2 Sensor Monitoring Node Code: #include

```

<WiFi.h>

const char* ssid = "AgriMind_Node"; const
char* password = "12345678";

WiFiServer server(80);

void setup() {
    Serial.begin(115200);

    WiFi.softAP(ssid, password);
    Serial.println("Sensor Node Started");
    Serial.println(WiFi.softAPIP());

    server.begin();
}

void loop() {

```

```
WiFiClient client = server.available();

if (client) {
  client.println("HTTP/1.1 200 OK");
  client.println("Content-
type:text/html"); client.println();

  client.println("AGRIMIND Sensor Node Active");

  client.stop();
}
}
```

4.4 Block Diagram

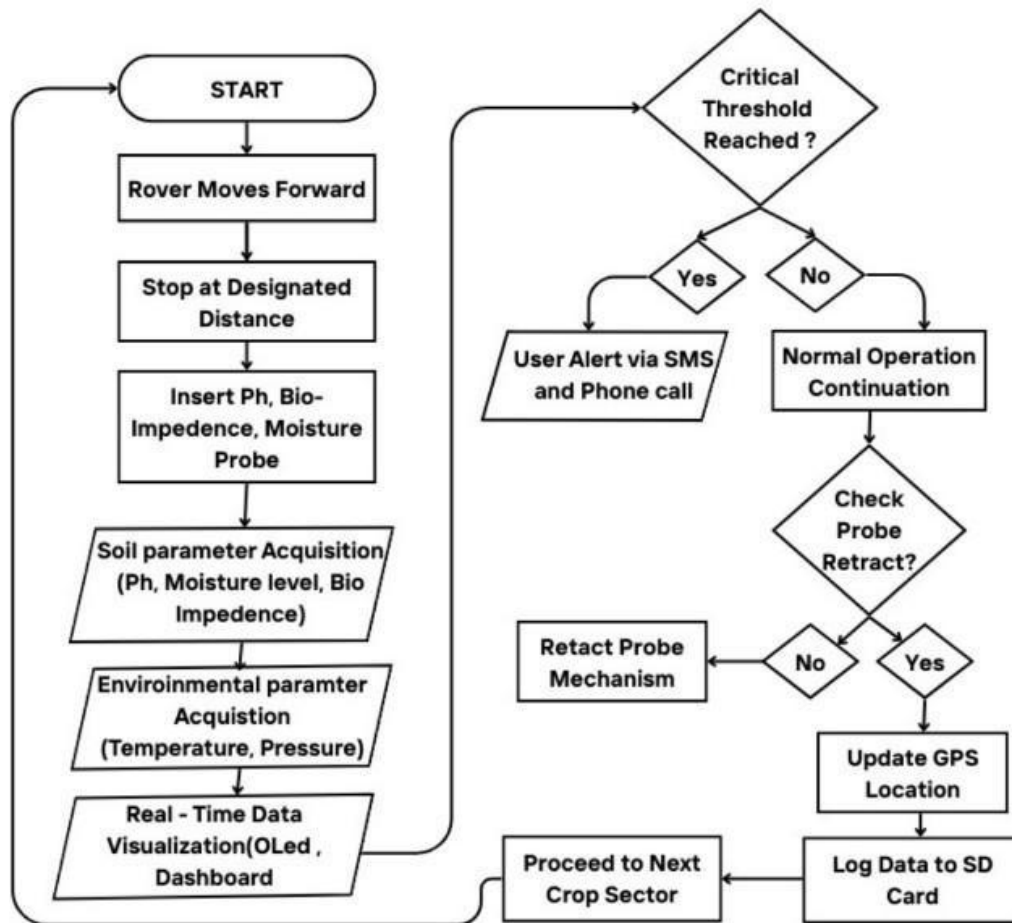


Fig 4.9 System Flow chart

CHAPTER V

RESULTS AND DISCUSSION

The AGRIMIND soil monitoring robot was implemented and successfully tested in different environments to assess its functioning. The robot proved itself capable of integrating sensors, the microcontroller, and communication devices successfully. The capacitive soil moisture sensor gave reliable results regarding the amount of soil moisture, showing differences between dry soil, medium soil moisture content, and moist soil. The pH sensor allowed one to conduct accurate measurement of soil acidity or alkalinity after calibration and perform simple soil analysis. The BMP280 sensor functioned successfully and gave precise measurements of temperature and atmospheric pressure.

Another factor evaluated concerning the rover is its mobility on varied surfaces such as dry ground and muddy ground. The L298N motor driver effectively facilitated the movement of the rover, with capabilities for moving forward, stopping, and changing direction. The installation of lugged wheels enhanced traction and made movement on rough and wet surfaces easier. Additionally, the ultrasonic sensor played a critical role in obstacle detection by recognizing obstacles on the path of the rover.

Wi-Fi communication provided real-time communication of the collected data from the sensors to the external device. There was successful establishment of the network connection, and the ESP32 was capable of transmitting data instantaneously within the range of its connectivity. Communication was achieved by remotely communicating the data, allowing the user convenience while retrieving soil information since the user did not have to be at the field location. There were some delays experienced during the transmission of the data, but these were very negligible.

The system performance proved to be efficient since there was constant updating of data from the sensors and fast decision-making on the basis of certain criteria. The robot behaved properly in response to variations in the soil moisture content. The system had low latency, thus making it ideal for real-time monitoring in agriculture.

But there are some constraints that were found during the process of implementation of this technology. First of all, the functioning of a pH sensor is dependent on its calibration. Secondly, the range of WiFi connection is quite restricted, which will pose problems for its use on large agricultural land plots. Thirdly, some power loss occurs because of overheating in the case of an L298N motor driver. Not with standing these challenges, the AGRIMIND system has shown that embedded systems and IoT can be effectively used in precision agriculture. This project represents a real-life implementation of the technology that will help farmers monitor soil parameters and optimize water usage. The modular nature of the system makes it possible to improve the functionality of the platform in the future by adding GPS for navigation, using cloud services for analytics, and implementing AI algorithms. **5.1**

Prototype

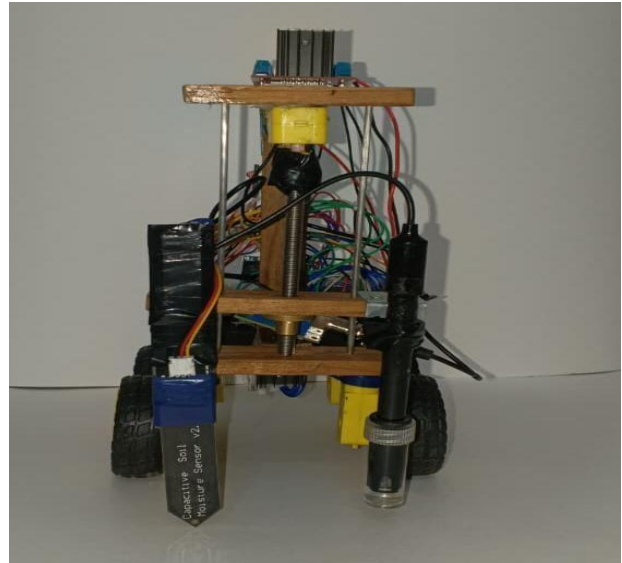
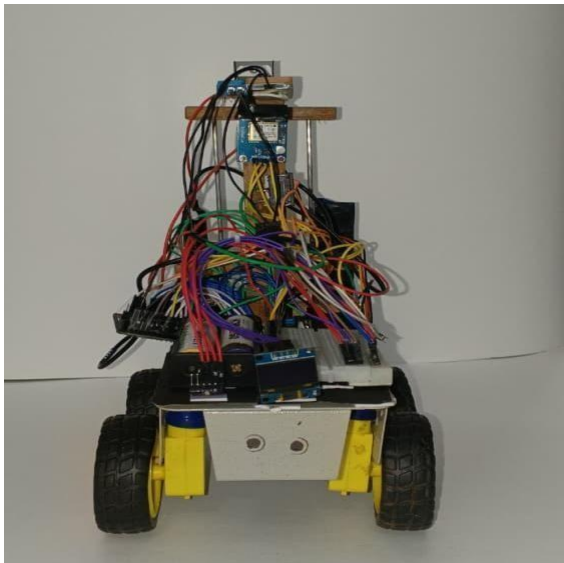


Fig 5.1 Prototype

CHAPTER VI

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION:

The AGRIMIND project provides an excellent example of how embedded and IoT devices can be utilized for modern agriculture monitoring purposes. The project aimed to monitor various soil and environmental indicators, including soil humidity levels, pH values, temperature, and atmospheric pressure. It is possible to acquire, process, and transfer all of the data due to the implementation of the ESP32 controller together with the use of various sensors. Additionally, since the data can be gathered through a rover that has mobility features, it is more efficient than a static system.

From the outcomes of implementation, it is clear that the system can provide reliable monitoring in varied environments. The moisture sensor detects the need for irrigation, while the pH sensor is able to offer information on the state of the soil. The BMP280 sensor plays an important role in environmental monitoring, while the motor control helps ensure that the rover moves smoothly. The integration of the Wi-Fi communication technology makes the system easy to use by allowing users to have access to data in real time.

Although the system is quite effective, it does have some limitations in terms of design. First, some of the sensors need calibration since their effectiveness varies with time. Second, the communication range is restricted by the Wi-Fi connectivity range. This means that for a big farm, this range would not cover all the area in the field. Finally, the motor driver utilized in the system has average efficiency, which results in power losses.

The future enhancement of this system will make it much more efficient. For example, by using more advanced communication methods like LoRa technology, the communication distance could be increased for big farms. In addition, including GPS technology will enable the localization and automatic navigation of the rover. Utilizing cloud platforms will make it possible to store data for a long time and carry out additional analytics. Finally, the implementation of machine learning algorithms would make it possible to conduct predictive analysis of soil quality and automate decisions regarding irrigation and fertilization.

In conclusion, the AGRIMIND soil monitoring rover provides a practical and effective approach to modern farming by combining sensing, automation, and communication technologies. The project

highlights the potential of IoT-based systems in improving agricultural productivity, optimizing resource usage, and supporting sustainable farming practices.

6.2 FUTURE ENHANCEMENTS:

AGRIMIND Soil Monitoring Rover System can be further enhanced using modern technology for the purpose of improving its efficiency and effectiveness. Among the improvements, one can include long-distance communication technology, including LoRa, which would overcome the limitations of WiFi technology in transmitting information over vast stretches of land. Such a change would render the system ideal for practical applications on vast agricultural lands.

Adding the GPS module would make it possible for the tracking of location in real time while providing the possibility for autonomous navigation as well. Using the right algorithm for determining the path, the rover can operate on its own along pre-defined paths. Additionally, connecting the device to the cloud will ensure that huge volumes of data are collected and stored for future purposes of analysis.

An important improvement is the inclusion of the ESP32-CAM board that would allow the farmer to monitor his/her crop visually in real-time. Visual information from photographs or videos could show whether plants have grown properly and whether their surrounding environment is healthy or not. It may be used for object detection via machine learning technology, where plants suffering from disease, presence of weeds, or any obstacle in the farm could be detected automatically.

Other features include the application of Vehicle to Vehicle Communication technology for exchange of data between the rover and an aerial drone. The vehicle to vehicle technology could help coordinate the activities of both the aerial drone and the ground rover by enabling them to exchange data among themselves. The drone would help collect information from above the field while the rover collects data on ground level.

Hardware efficiency may be improved by changing the existing inefficient L298N driver to an efficient one to minimize electrical power wastage and extend the duration of the battery. Energy efficiency in the system can also be enhanced by adopting an alternate source of energy such as solar energy. Improved sensor accuracy and installation of additional sensors will make the system reliable.

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